LESSON 10.4

Math is Ingenuity

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use ratios and models to explain how the Penny Farthing and gear-driven bicycles were ingenious solutions to a real problem.

WHY THIS MATTERS The invention of the bicycle in the late 1800s offered people a way to move about easily. Among the first bicycles created was the Penny Farthing, with a large front wheel and a small rear wheel. Ingenuity in mathematics helps us come up with different, and sometimes unique, solutions to problems.

REMEMBER

Math is Ingenuity

Gear Ratios, Circumference & Mechanical Advantage. Formula: Distance = Rotations × Circumference | Gear Ratio = Teeth_Driver / Teeth_Driven The Penny Farthing is powered by the rider: pushing the pedals rotates the front wheel directly, so each pedal rotation rotates the wheel one time — a 1 to 1 ratio.

1. Circumference is the distance traveled in one full wheel rotation.
2. Gear ratios determine speed and rotational force (torque).
3. Apply proportional reasoning to calculate travel distance and mechanical advantage

Words: ingenuity *(ingenio)* · rotation *(rotación)* · gear *(engranaje)* · ratio *(razón)* · equivalent ratios *(razones equivalentes)*

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each rear gear by how the wheel responds to one pedal turn.

Groups: More than one wheel rotation · One rotation or less

_____ 12-tooth rear gear

_____ 16-tooth rear gear

_____ 24-tooth rear gear

_____ 48-tooth rear gear

_____ 60-tooth rear gear

_____ 96-tooth rear gear

HINT Gear ratio = driver teeth ÷ driven teeth = 48 ÷ rear teeth.

2 **Complete the table. Show how you found each value.**

The pedal gear has 48 teeth. Complete the table of wheel rotations per pedal turn.

Rear gear (teeth)	Ratio 48 ÷ teeth	Wheel rotations per pedal turn
12		
16		
24		

HINT Divide the driver's 48 teeth by the rear gear's teeth.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

In the Tower of Hanoi, what is the fewest moves needed for 3 disks?

(A) 3
(B) 6
(C) 7
(D) 8

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Look at the pattern 1, 3, 7, 15. What is the rule?

(A) Add 2 each time
(B) Double the term
(C) Add the two terms before it
(D) Double the term and add 1

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

What is the next term in 1, 3, 7, 15, ___?

(A) 17
(B) 23
(C) 30
(D) 31

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Your partner says the bigger rear gear makes the wheel turn more. Is that right, and what stays equivalent in the ratio of teeth when you switch gears?

SENTENCE STARTER With the ___-tooth gear the ratio is ___, so the wheel ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 10.4

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which statement best captures what this lesson means by 'Math is Ingenuity'?

(A) Math can help us come up with creative solutions to everyday problems

(B) Math is only useful for inventors who lived in the 1800s

(C) Ingenuity means avoiding math and just guessing

(D) Every bicycle must have a 1 to 1 pedal ratio

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

If 3 disks take 7 moves, how many moves do 4 disks take?

(A) 10

(B) 11

(C) 14

(D) 15

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Humans have long looked for ways to travel easily and quickly. One modern option is a motorized unicycle. Imagine what a future mode of individual transportation might look like. A future vehicle's design uses a motor gear with 40 teeth turning a wheel gear with 10 teeth. Following the gear-ratio math from this lesson, how many wheel rotations happen for each motor gear rotation?

(A) 0.25 rotations

(B) 4 rotations

(C) 30 rotations

(D) 400 rotations

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Humans have long looked for ways to travel easily and quickly. One modern option is a motorized unicycle. Imagine what a future mode of individual transportation might look like. Why does a complete connect answer need to name a specific piece of math instead of just saying 'it uses math'?

(A) Because grading requires exactly one number in every answer

(B) Because the word 'math' cannot be used in a written answer

(C) Naming the actual ratio, measurement, or rate shows the math really shaped the design, not just that math exists somewhere

(D) Because a vehicle description never needs any math at all

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Humans have long looked for ways to travel easily and quickly. One modern option is a motorized unicycle. Imagine what a future mode of individual transportation might look like. A student's answer says only, 'My vehicle is powered by a battery.' Which addition would make this a complete connect answer, based on this lesson's check?

(A) Saying the vehicle looks very futuristic

(B) Saying the vehicle is extremely fast

(C) Repeating that it is powered by a battery a second time

(D) Naming a specific ratio or rate in the design, like a gear ratio between a motor and a wheel

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Your partner says the bigger rear gear makes the wheel turn more. Is that right, and what stays equivalent in the ratio of teeth when you switch gears?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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